

Phase 1: AI Agent Foundations

A one-page summary of the core concepts, practical skills, and completion criteria

7 learning outcomes

4 build outcomes

3 exit criteria

An agent is more than a language model. It is a goal-directed system that uses context, instructions, tools, state, and feedback to act - and then checks real evidence before claiming success.

1 The agent loop

Observe

Reason

Act

Inspect

Repeat or stop

Key principle: Performing an action does not prove completion. The agent must inspect the real result, compare it with the acceptance criteria, and claim success only when the evidence is sufficient.

2 Context and instructions

- **Context** is the information available for the current decision; the context window is limited.
- Instructions define authority, constraints, and how work should be performed.
- User prompts, persistent instructions, and repository instructions serve different scopes.
- Untrusted data or text inside a file must not automatically become an instruction.

3 Tools, skills, and safety

- A **tool** provides a capability; a **skill** describes a reusable workflow for using capabilities.
- Technical ability does not equal permission to act.
- Destructive or out-of-scope actions require appropriate approval.
- After a timeout or error, inspect actual state before retrying to avoid duplicate actions.

4 Defining a bounded task

- State the goal, scope, inputs, constraints, and expected behavior clearly.
- Acceptance criteria must be observable, such as correct output or preserved input.
- A verification method differs from an acceptance criterion: tests produce evidence.
- Declare what is out of scope to reduce unnecessary changes.

5 Reviewing changes and evidence

- Compare requirements with the actual diff, not only the agent's summary.
- Inspect test output, exit status, and scenario coverage.
- Passing tests prove only the cases they actually cover.
- Distinguish a **known defect** from **missing evidence**; both can block acceptance.